

For the second scenario...

- The same for the 1st student, the 2nd student and the 3rd student (because the 2nd student can guess the 1st student's draw and the 3rd student can guess both the 1st student's draw and the 2nd student's draw).
- Suppose R, R, R. Note that the 4th student knows that the third R is not informative as it does not reflect the 3rd student's draw. Intuitively, the 4th student is exactly in the same situation as the 3rd student.
- The same for the 5th student after R, R, R, R....

Conditional Probability and Bayes' Rule

What is the probability that the urn is RED after many draws of red balls and blue balls in either scenario?

We introduce the notion of **conditional probability** and a useful formula called **Bayes' rule**.

Let X and Y be two possible events. $\Pr[Y|X]$ is the **conditional probability** of event Y given that X occurs.

For example, imagine throwing a dice. X can be "even numbers" and Y can be "6". Then $\Pr[Y|X]$ would be $1/3$.

Conditional Probability and Bayes' Rule

$\Pr[Y|X]$ is defined as follows:

Conditional Probability

$$\Pr[Y|X] = \frac{\Pr[X \cap Y]}{\Pr[X]}$$

, where $\Pr[X \cap Y]$ is the probability that both X and Y occurs.

Similarly $\Pr[X|Y] = \frac{\Pr[X \cap Y]}{\Pr[Y]}$. By combining them, we have the following useful formula that connects two conditional probabilities $\Pr[Y|X]$ and $\Pr[X|Y]$:

Bayes' Rule

$$\Pr[X|Y] = \frac{\Pr[X] \Pr[Y|X]}{\Pr[Y]}$$

- In the following, we consider a setting where there is some hidden state θ in $\{\theta_1, \dots, \theta_n\}$ that follows some distribution and there is a informative signal s in $\{s_1, \dots, s_m\}$ that follows some conditional distribution $\Pr[s = s_j | \theta = \theta_i]$.
- We are interested in conditional probability of some state given some signal realizations such as

$$\Pr[\theta = \theta_i | s = s_j] = \frac{\Pr[\theta = \theta_i] \Pr[s = s_j | \theta = \theta_i]}{\Pr[s = s_j]}$$

We call $\Pr[\theta = \theta_i]$ **prior belief** on θ_i and $\Pr[\theta = \theta_i | s = s_j]$ **posterior belief** on θ_i after observing s_j .

- One useful fact (verify this): $\Pr[\theta = \theta_i] = \sum_{j=1}^m \Pr[s = s_j] \Pr[\theta = \theta_i | s = s_j]$.
i.e. the expected value of the posterior belief = the prior belief.

Let's apply Bayes' rule to the experiment.

- The conditional probability of RED given one draw of red is

$$\frac{\Pr[\text{the urn is RED and one red draw}]}{\Pr[\text{one red draw}]} = \frac{1/2 \times 2/3}{1/2 \times 2/3 + 1/2 \times 1/3} = 2/3.$$

- The conditional probability of RED given two draws of red is

$$\frac{\Pr[\text{the urn is RED and two red draws}]}{\Pr[\text{two red draws}]} = \frac{1/2 \times (2/3)^2}{1/2 \times (2/3)^2 + 1/2 \times (1/3)^2} = 4/5.$$

- More generally, the conditional probability of RED given m draws of red and ℓ draws of blue is

$$\frac{1/2(2/3)^m(1/3)^\ell}{1/2(2/3)^m(1/3)^\ell + 1/2(1/3)^m(2/3)^\ell}.$$

(Note that the order of red draws and blue draws do not matter.)

- For the second scenario, the conditional probability of RED for the 3rd student after R, R and drawing r is

$$\frac{\Pr[\text{the urn is RED and } (R, R, r)]}{\Pr[R, R, r]} = \frac{\Pr[\text{the urn is RED and 1, 2, 3 draw red}]}{\Pr[1, 2, 3 \text{ draw red}]}$$

- The conditional probability of RED for the 4th students after R, R, R and drawing r is $\frac{\Pr[\text{the urn is RED and } (R, R, R, r)]}{\Pr[R, R, R, r]} = \frac{\Pr[\text{the urn is RED and 1, 2, 4 draw red}]}{\Pr[1, 2, 4 \text{ draw red}]}$ which is exactly the same as the above conditional probability for the 3rd student.

Another exercise of Bayes rule,

- 1% of population is infected by some disease.
- There is a testing to detect infection: For an infected, the test returns a positive 99%, but returns a false negative 1% of the time. For a healthy person, the test returns a negative 99%, but returns a false positive 1% of the time.
- Suppose that you test a random person and gets a positive. What is the conditional probability that the person is infected?

Sequential Decision Problem

We study the following sequential decision problem and examine if and how people sharing information converge to the same choice.

- There are two restaurants. One is old and well known. The value of its meal is normalized to 0. The other one just opened recently, so people do not know how good it is. If it's good, the value of each meal is 1. If it's bad, the value of each meal is -1 . Assume that the restaurant is either a good restaurant or a bad restaurant with equal probability.
- Each period, a person observes a signal $s \in \{g, b\}$, which is correct with probability $2/3$, then pick one restaurant to have a lunch.
- Each person can obtain some information from the all preceding customers (we can think of this as information network).

Observational Learning and Information Cascade

- Suppose that each person can observe all the signals of the preceding customers. So, the n th person observes $n - 1$ independent signals about the quality of the new restaurant. Let $x(n) \in \{old, new\}$ be the choice of the n th person.
- It is a well-known result in Statistics that the sequence of posterior beliefs converge to put probability 1 to the true state as $n \rightarrow \infty$ with probability 1. As a consequence, they make a right choice with high probability for large n . To be precise, $\Pr[x(n) = new|good] \rightarrow 1$ and $\Pr[x(n) = new|bad] \rightarrow 0$ as $n \rightarrow \infty$.

- Now suppose that each person does not learn others' private signals directly, but observes the restaurant choices of all the preceding customers. This is a model of **observational (social) learning**.
- Then each person makes a choice as follows.
 - ▶ The 1st and 2nd person make a choice based on their private signal.
 - ▶ If the 1st and the 2nd person make a different choice, then the 3rd person makes a choice based on her private signal. If they make the same choice, then the 3rd person make the same choice independent of her signal. Once this happens, then the 4th, the 5th,... all subsequent people make the same choice without using their information (**Information Cascade**).

- More generally, once $(2k - 1)$ th person and the $(2k)$ th person choose the same restaurant, then all subsequent people choose the same restaurant.
- By the end of the second period, they converge to the right choice with probability $4/9$, the wrong choice with probability $1/9$, and come back to the starting point with belief $1/2$ with probability $4/9$. Eventually, they converge to the same action with probability 1 but converge to the wrong one 20% of the time (why?).
- When a new restaurant is chosen consecutively and the information cascade starts, their posterior belief that the new restaurant is good is $4/5$. The posterior belief of everyone after this event is either $8/9$ (after observing g) or $2/3$ (after observing b).

What are takeaways from this model?

- People may converge to the same choice when they learn from others. In fact, they do so with almost probability 1 when there are many people.
- When people learn from others' behavior, they may end up converging to a wrong choice with positive probability however large n is.
- A cascade can be fragile. More and more people choosing the same action does not mean that they are getting more and more confident about their choice.

Comment.

- This is a very simple model to make a particular point (about information cascade), so there are many unsatisfactory parts.
 - ▶ Information sharing network is very special. For example, each person may observe only a subset of the actions of the predecessors.
 - ▶ Each person may care not only about the true state but also about other people's choices.
 - ▶ Each person may observe more than one signal.
 - ▶ People may have different prior beliefs and the accuracy of their signals may be different.
 - ▶ People may not have a precise knowledge of information network to apply Bayes' rule correctly.
 - ▶ ...

Selective Information Sharing

There are many variations of the basic Bayesian learning model to address various issues.

Among them, we study a model with selective information sharing and misperception of the information dynamics.

Consider the following model of selective information sharing (Bowen et.al. 2023)

- There is a hidden true state, which is either A or B with equal probability.
- There are three types of people: normal, A -dogmatic, B -dogmatic. There is a general undirected information network that connects them.
- In period $t = 1, 2, \dots$, person i observes nothing with probability $1 - \gamma$ and a private signal $s_{it} \in \{a, b\}$ with probability $\gamma \in (0, 1)$, which is correct with probability $q \in (0.5, 1)$. The signals are i.i.d. across people and time.
- A normal person shares all private signals, a A -dogmatic person only shares a signals, and a B -dogmatic person only shares b signals with all neighbors.
- **Misperception:** Everyone believes that the probability of observing an informative private signal is $\hat{\gamma} \in (0, \gamma)$.

Comments.

- You cannot fake a signal, but you can hide it.
- If you have more A -dogmatic friends than B -dogmatic friends, then you tend to observe more a signals. However, each (normal) person is aware of it. So, biased friends do not necessarily cloud your judgment.
- We find that it is the misperception of γ ($\hat{\gamma} \in (0, \gamma)$) that generates a certain type of bias: **polarization**. Intuitively, when γ is large, you infer from a silence of your A -dogmatic friend that she is hiding an inconvenient signal. A smaller $\hat{\gamma}$ reduces this effect.

Polarization

Consider a normal person i whose only friend is an A -dogmatic friend j . Let's see how her belief would be updated in each period.

- Suppose that she observes a and learns a from her friend. The posterior belief on A is

$$\Pr[A|s_i = a, s_j = a] = \frac{1/2\hat{\gamma}^2 q^2}{1/2\hat{\gamma}^2 q^2 + 1/2\hat{\gamma}^2(1-q)^2} = \frac{q^2}{q^2 + (1-q)^2}$$

This is independent of $\hat{\gamma}$.

- Suppose that she observes a and learns nothing (n) from her friend. The posterior belief on A is

$$\begin{aligned}\Pr[A|s_i = a, s_j = n] &= \frac{1/2\hat{\gamma}q((1-\hat{\gamma}) + \hat{\gamma}(1-q))}{1/2\hat{\gamma}q((1-\hat{\gamma}) + \hat{\gamma}(1-q)) + 1/2\hat{\gamma}(1-q)((1-\hat{\gamma}) + \hat{\gamma}q)} \\ &= \frac{1}{1 + \frac{1-q}{q} \frac{(1-\hat{\gamma}) + \hat{\gamma}q}{(1-\hat{\gamma}) + \hat{\gamma}(1-q)}}\end{aligned}$$

This would be **smaller** with true $\gamma > \hat{\gamma}$.

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- This inflated belief on A realizes not only for $s_i = a$ but also for $s_i = b$, n each time your A -dogmatic friend is silent.
- We know that, without any misperception, the expected posterior belief on A is equal to the prior belief on A , which is $1/2$. Here the expected posterior belief on A (expectation with true γ) must be larger than $1/2$ with $\hat{\gamma} < \gamma$.
- As this is repeated in period $t = 1, 2, 3, \dots$, we can show that each normal type's posterior belief converges to A with probability 1, irrespective of the true state.
- Similarly, any normal person with only one B -dogmatic friend believes that the state is B eventually with probability 1. So, we have a polarization in beliefs.

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A general result for more general network is as follows:

Suppose that a normal person has d_N normal friends, d_A A -dogmatic friends, and d_B B -dogmatic friends with $d_A > d_B$. Her friends share their signals, and she updates her belief based on their signals with $\hat{\gamma} < \gamma$ as described above. Then, given any prior belief on $\pi \in (0, 1)$ on A , her posterior belief on A converges to 1 with probability 1 if the private signal is not too informative (i.e. q is in some middle range $q \in (0.5, \bar{q})$).

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Summary

- We learned various models of Bayesian social learning.
- If a person observes i.i.d. signals about the state, she eventually learns the true state with probability 1.
- In a model of observational learning where people learn only from others' behavior:
 - People end up choosing the same action without utilizing their information.
 - And they can be wrong.
 - A larger herd does not imply a stronger confidence about their choice. So, herding is fragile.

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Summary (continued)

- We studied a variation of Bayesian learning model with selective information sharing and misperception, and derived a polarization in beliefs.

Suggested reading: EK Ch.16